

Event-Connect: Universal Multi-API Hub

Serverless Event Registration & Ticketing Integration System on AWS

SUBMISSION LINKS & ENDPOINTS:

GitHub Repository: <https://github.com/GeekKwame/event-registration-system-sam>

Live Web App (GitHub Pages) <https://geekkwame.github.io/event-registration-system-sam/>

Primary AWS API URL: <https://kems8drwn6.execute-api.us-west-1.amazonaws.com/Prod>

1. Executive Summary & Problem Statement

Event-Connect replaces manual Microsoft Forms + Excel spreadsheets with a scalable AWS serverless REST API and web dashboard. The system automates capacity control, eliminates duplicate registrations, dispatches instant SNS confirmation emails, and connects to multiple student API Gateways across regions.

2. AWS Architecture & Components

- AWS API Gateway: REST API router with CORS support and stage normalization.
- AWS Lambda (Python 3.12): 4 microservices for listing events, registering, lookups, and cancellations.
- Amazon DynamoDB: EventsTable and RegistrationsTable with EmailIndex GSI.
- Amazon SNS: Confirmation topic publishing instant email receipts.
- Amazon CloudWatch: 14-day log group retention + RegisterErrorRateAlarm (>5% threshold).
- GitHub Actions CI/CD: Automated testing on PR + SAM deployment on push to main.

3. Deliverables Verification Matrix (PDF Page 7)

#	Required Deliverable	Implementation / Path	Status
1	GitHub Repository	github.com/GeekKwame/event-registration-system-sam	PASSED
2	CI/CD Pipeline	.github/workflows/deploy.yml & pages.yml	PASSED
3	Lambda Functions	src/handlers/ (list, register, get_regs, cancel)	PASSED
4	DynamoDB Tables	template.yaml (Events & Registrations + GSI)	PASSED
5	CloudWatch Alarms	template.yaml (RegisterErrorRateAlarm >5%)	PASSED
6	README File	README.md (Architecture, Setup & Live Endpoints)	PASSED
7	Product Presentation	docs/PROJECT_PRESENTATION.md & Submission PDF	PASSED